



Warehouses Personnel Auxiliary services

Tompkins Ch4,7,9 and Stephens Ch8,9,12

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Ch7: Warehouse functions

- Sometimes it is difficult to understand whose responsibility it is...operations? Logistics? Operation excellence? Purchasing/procurement? Or IT?
- Sometimes a company's core business = warehousing (Take-a-Lot, DHL,DSV)
- “planning and management of warehouses are very challenging – we must turn to *simplification* and *process improvement*...”
- Functions:
 - Receiving of materials
 - Inspection and quality control
 - Repackaging as needed
 - Categorization
 - Putaway to Storage
 - Order picking
 - Sorting
 - Packing and shipping of orders
- Warehouse design/Inventory management are subjects in their own right

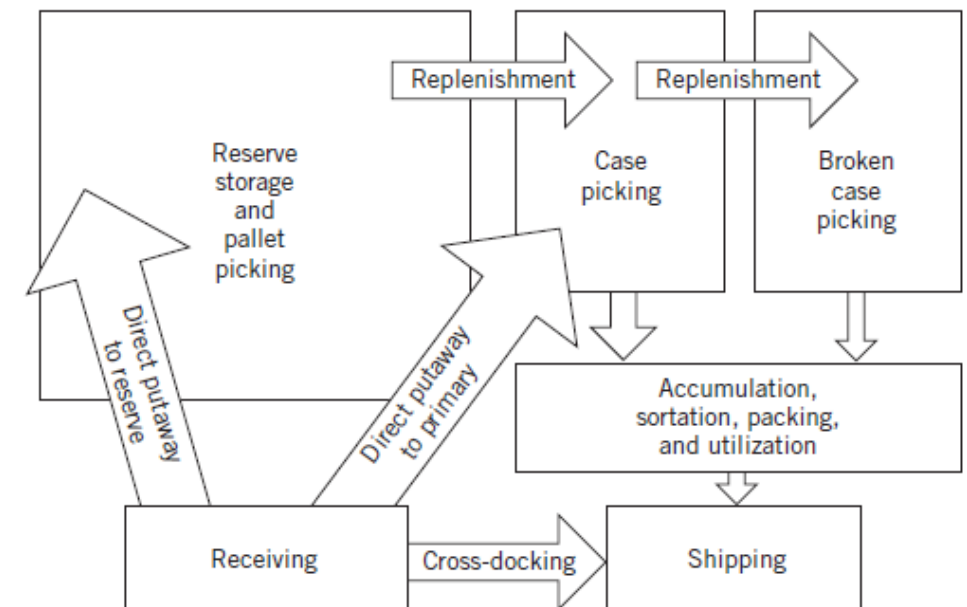


Figure 7.2 Typical warehouse functions and flows.

Ch7: Warehouse context

Large warehouses often include RECEIVING and SHIPPING functions (p392)

However, distinguish between warehousing and distribution centres

Facility requirements can include: (p392)

- Process start the moment the carriers cross the PROPERTY line
- Security and access policies
- Docking stations
- Dock levelers
- Loading/unloading methods
- Sufficient staging and preparation areas
- Technology integration to systems
- Safe effective material handling

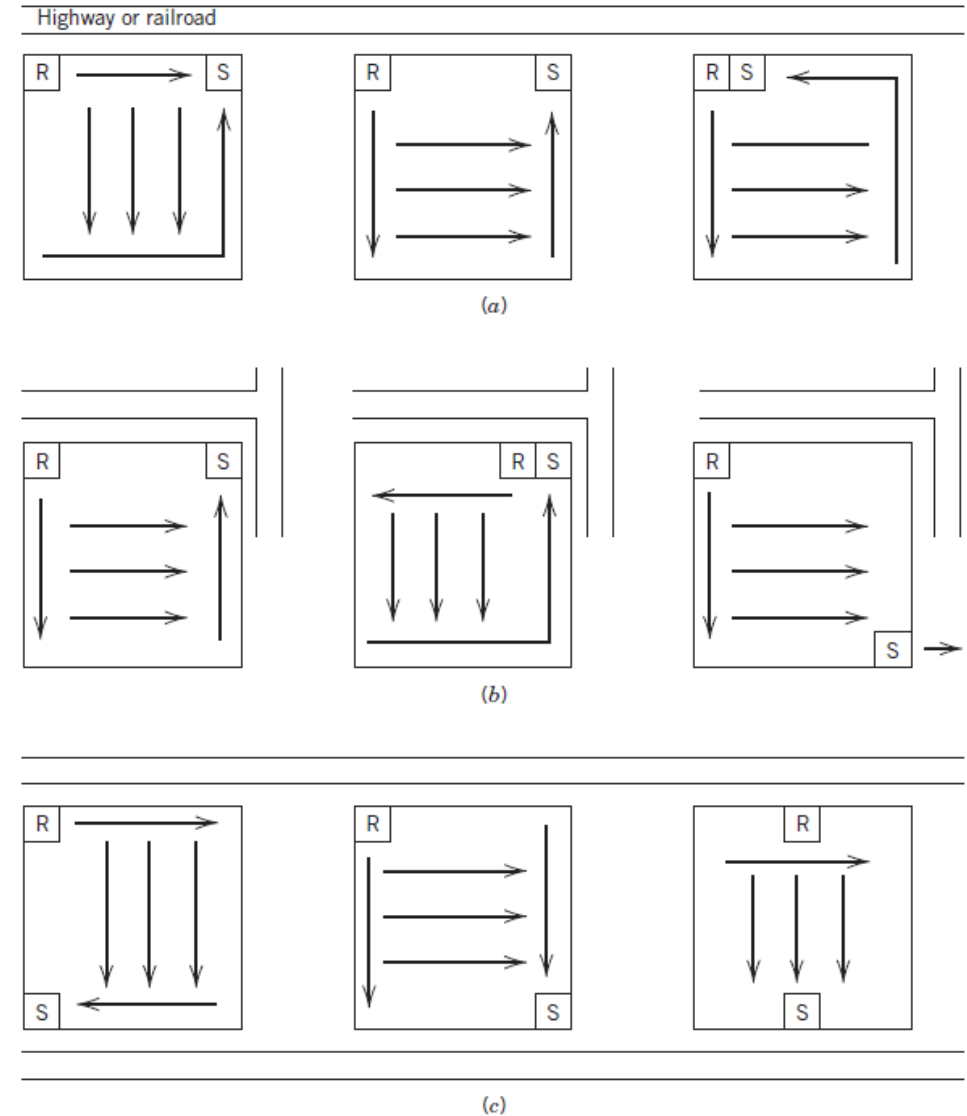


Figure 7.3 Possible arrangements of shipping and receiving areas. (a) Transportation facilities on one side of building. (b) Transportation facilities on two adjacent sides of building. (c) Transportation facilities on two opposite sides of building. (With permission from Apple [1].)

Ch7: Warehouse space planning

stuff must be stored such that it can be FOUND

Steps to determine SPACE requirements p399)

- *Step1 Determine WHAT, HOW MUCH and WHEN stuff is to be received and shipped (fig 7.4)*

Company _____ Date _____ Raw Materials _____ Finished Goods _____
Prepared by _____ Sheet _____ of _____ Plant Supplies _____

[illegible]

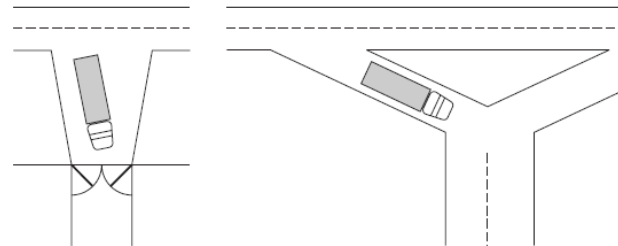
Figure 7.4 Receiving and shipping analysis chart.

Ch7: Warehouse space planning

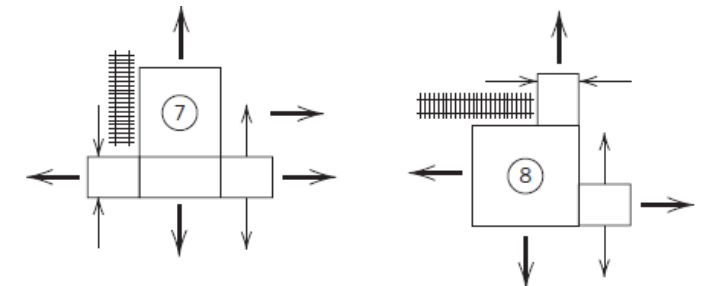
- *Step2 Determine the NUMBER and TYPE of 'docks' (waiting line analyses, historical data, work sampling, time studies. Sometimes there are standards e.g. fig 7.5)*

“considerable effort” is required to obtain the information...

- *Step3 Determine space for In/Out WITHIN facility (remember the 'holding' area as well as the need for maneuvering of transport)*



- Tips on Truck guidelines p401-406
- Include convenience/waiting areas, trash disposal, pallet and packing material storage, trucker's lounge (share silos story), buffer/staging areas, material handling equipment storage and movement areas
- Dock operations (levelers, locations, expansions) guidelines p407-415



Ch7: Storage space planning template

Space		Unit Loads		Quantity of Unit Loads Stores				Storage			
Description (1)	Type (2)	Capacity (3)	Size (4)	Weight (5)	Maximum (6)	Average (7)	Planned (8)	Method (9)	Specs (10)	Area (ft ²) (11)	Ceiling Height Required (ft) (12)
Aluminum rails, three runner 288"	Bundles	50 pcs.	18" × 28" × 288"	1250 lbs.	14	5	12	Cantilever rack	Four-arm dual rack, 4' × 12' × 6'	192	24
Glass, 1/4" thick, 8' × 4' sheets	Racks	4 sheets	8' × 4' × 4'	400 lbs.	20	13	15	Pallet rack	4' × 4' × 22' (4 levels)	240	24
Rubber stripping 1/8" square	Cartons	500 ft	12" × 4" × 12"	20 lbs.	20	12	8	Industrial shelving	5 level 42" × 18" shelves	12	9
Adhesive, water sealant	Drum	350 lbs.	2' diameter × 4'	375 lbs.	7	3	6	Floor simple	2.5' × 2.0' floor loaded	60	8
Packing boxes 2' × 4' × 4"	Pallets	500	40" × 48" × 48"	600 lbs.	10	8	6	Bulk stack	4' × 4' × 12' (3 levels)	80	15

Figure 7.16 Storage analysis chart.

Ch7: Warehouse operations

- Ex 7.1 – note all the detail specs that determines the calc
- Storage space planning (influenced by Inventory management policies)
- Ex 7.2 as it relates to Inventory management decisions
- Ex 7.4 ABC layout idea demonstrated (A most moving vs C least moving items)
- Storage layout planning according to principles (p424 NB)

Popularity/ Similarity/ Size/ Characteristics (eg perishable, oddly shaped, crushable, hazardous, security, compatibility eg chemicals)

- Fig 7.28 Proper and improper layout examples
- Concept of a unit load = NB design determining parameter
- Use 3D space as there is less movement
- Determination of aisles = key factor in warehouse design

Ch4 : Office sample

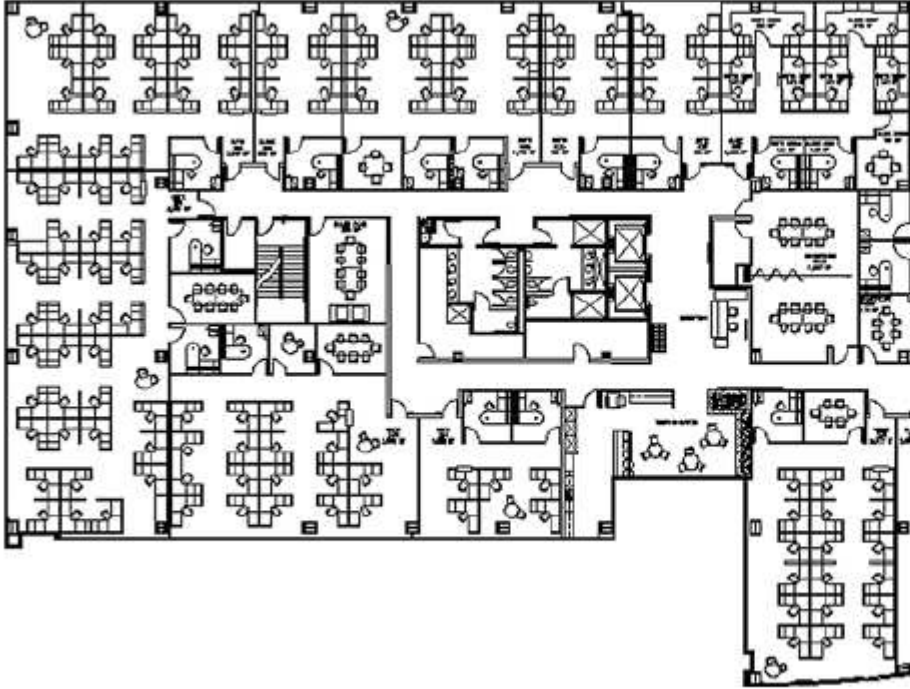


Figure 4.15 Floor plan of office areas. (Courtesy of)

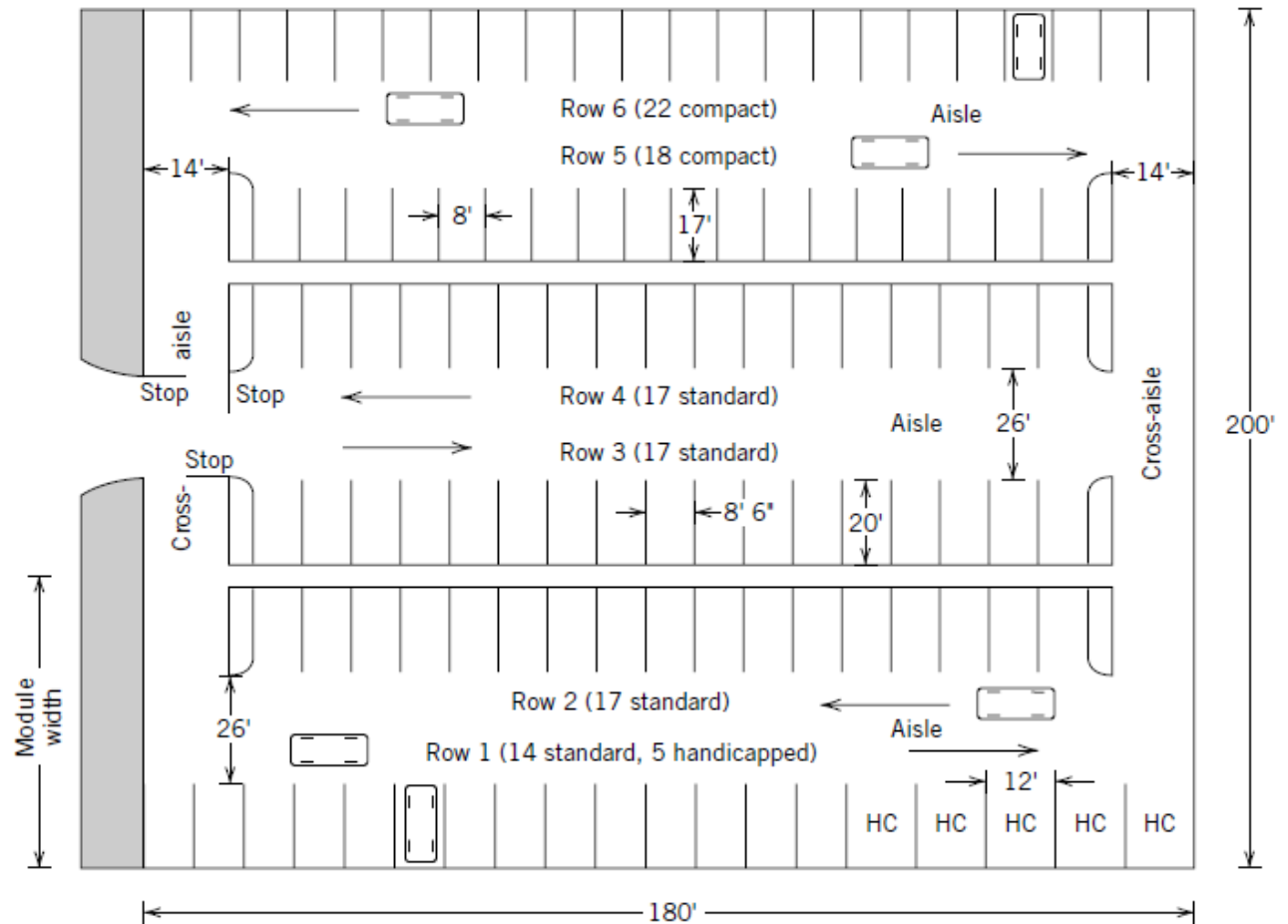


Figure 4.3 Parking lot for Example 4.1.

Ch4 : Personnel requirements include

- Parking (Fig4.1, 4.2 and Table4.1, Examples 4.1 and 4.2)
- Entrance/exit
- Lockers for storage of personal belongings (
- Restrooms (4.3)
- Food services (4.4)
- Health/medical services
- Recreational services
- Offices (4.7) (also SMART offices)

*Difficult to plan due as it is dependent on management philosophies
(see examples in book on how it can differ)*

Ch9: Facility auxiliary systems

- These are often architects, civil engineers & builder's responsibilities
- BUT the facility planner need to know WHAT systems are required WHERE
- Systems to include:
 - Structural (walls, columns, roofs) (9.2)
 - Enclosure (barriers against wind/water/humans incl: walls, windows, floors and for comfort incl: doors, ventilation, thermal) (9.3)
 - Atmospheric (temperature incl heating/cooling, humidity, clothing, "clean" rooms) (9.4)
 - Electrical (lighting and other systems requiring electricity) (9.5)
 - Life safety (fire protection, seismic events and power failures) (9.6)
 - Sanitation (water, plumbing) (9.7)
- The idea of Automation (think SMART)
- Sometimes Maintenance functions are also seen as part of auxiliary systems